

Discmath Week 2

Emil Straschil

A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.

Hello

I am Emil.

I have a **website**:

emils.site (yes thats the URL)

repo: <https://github.com/emil3tr/emil3tr.github.io>

All materials will be uploaded there.

Any Feedback / Questions /
Wishes / ... ?

→ estraschil@student.ethz.ch

→ “Emil” (floxi4) on dinfk-discord

→ Diskmath-questions: ask here
so others can benefit (:

Where are we right now?

Basics

Abstraction
Formulas
Statements
Prop. Logic
Pred. Logic
Proof Patterns

Sets and
Relations

Sets
Set Operations
Relations
Equivalence
Partial Order
Functions
Countability

Number Theory

Division
Primes
Modular Arith.
Diffie-Hellman

Algebra

Monoids
Groups
Euler Totient
RSA
Rings
Polynomials
Finite Fields
Err. Corr. Codes

Logic

Proof Systems
Logic
Calculi
Res. Calculus
Prop. Logic
Pred. Logic

Today

Hope you had a good start!

1. Questions?
2. Introduction
3. How to Study
4. Plan for Exercise Sessions
5. Last Exercise Sheet Review
6. Kahoot
7. Theory Recap
8. Exercises

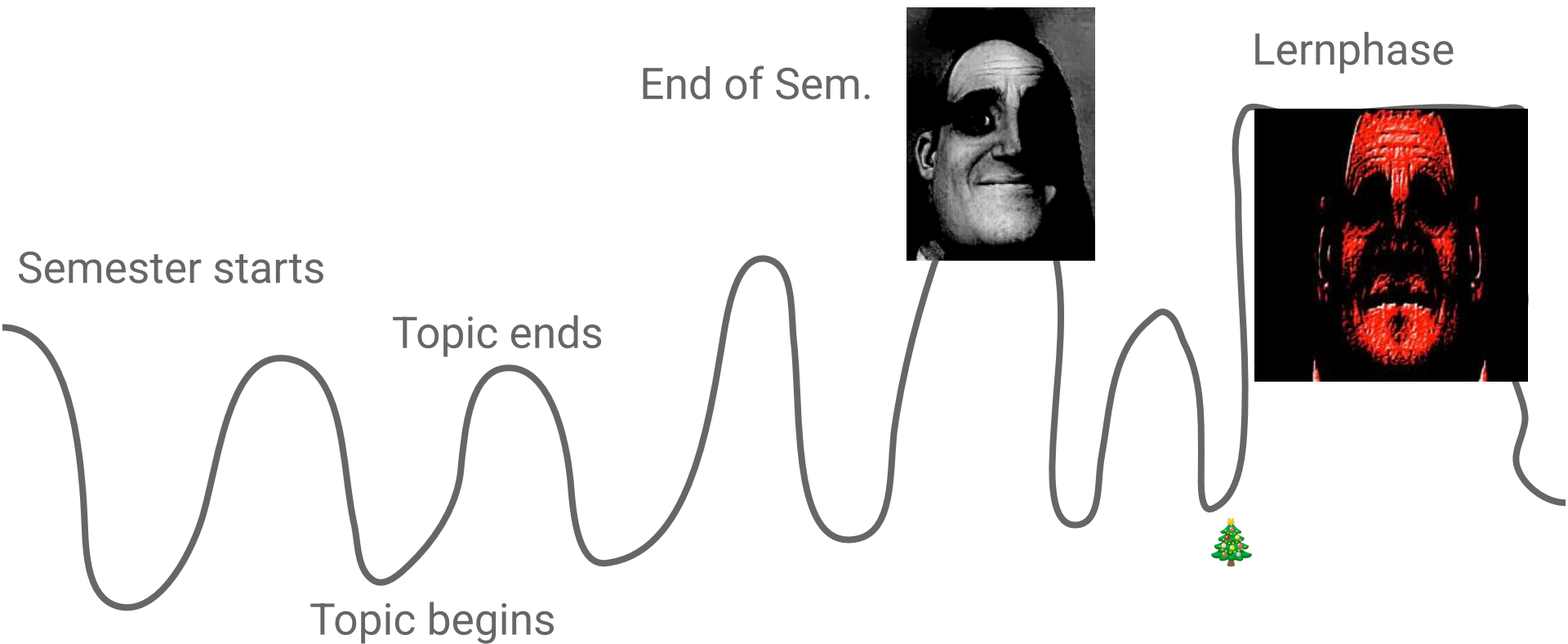
Questions?

About Me

- 3rd year student
- took DM 2 years ago
- was a TA last year

About Diskmath...

Diskmath Semester Stress Level



You are not stupid

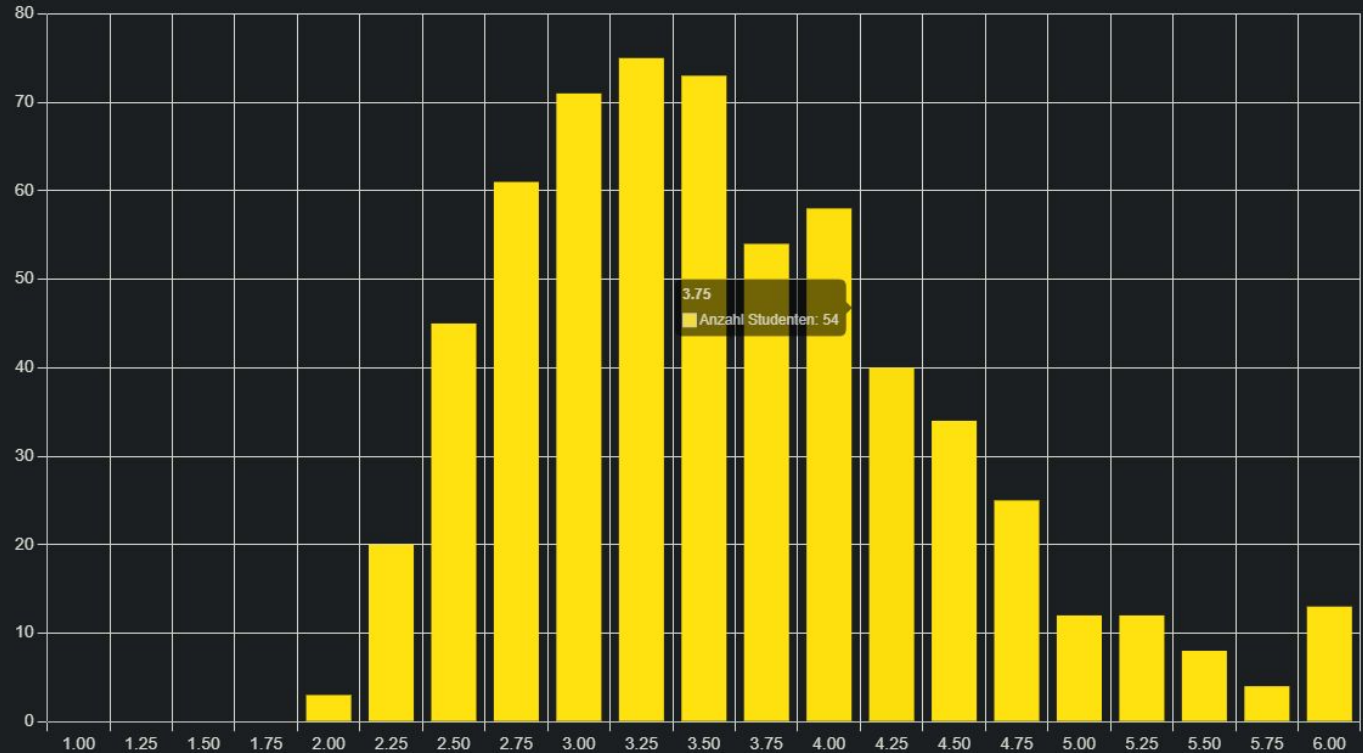
Grades:

passrate ~40%

pass = 30% on
exam

→ not knowing
= normal

→ ASK
QUESTIONS !!!



BUT: You CAN do it!

“This is difficult, so I cannot understand it.” = **WRONG**

BETTER: “This is actually not that hard, I just need to find a way to understand it.”

Diskmath is **NOT** complicated, everything is logical. Just difficult to understand at first.

One Week of Diskmath

Mo	Di	Mi	Do	Fr	Sa / So
Lecture	Session here Bonus?	Lecture		+Serie	
READ THE SCRIPT, discuss with colleagues					

ASK QUESTIONS!!

You need to prove stuff in the exam!

This requires understanding of that stuff.

Best way to gain deep level of understanding? → Ask and answer questions to / from colleagues.

Study Tips

The Obvious...

- go to the lectures
- go to the exercise classes
- do the homeworks
- read the script
- ask questions (:

It's your first semester, try to complete it "as intended". You will find your own rhythm after some time.

My Tips

- Read the script **before** the lecture
- Take a peek at old exams
- Discuss together
- But solve exercises alone
- Explain to your friends
- Explain to yourself
- Work on paper (or tablet)

My Tips

Use AI as little as possible.

Asking / Explaining to friends is much better.

You need to learn how to understand hard things yourself.

About Exercise Sheets

Try every exercise!

→ Stuck? Wait some days. Don't skip it, ask questions!

→ Really stuck? Skip it, don't waste time

→ ChatGPT can be helpful, but real humans are better

→ Look at the solutions

→ **!!!! Do it again ON PAPER !!!!** even if you saw the solution

Bonus Points

- Different from last year
- Two sessions per semester (look at moodle)
- Bring solution **on real paper** and **student id**
- Don't be scared, most TAs are nice
- We will ask you additional questions about the topic. (Know your definitions etc.)
- Be on time! (Most TAs have sessions afterwards.)

Questions?

Plan for my sessions

General

- My notes / slides / exercises are in english
- The script is amazing, read it!
- Otherwise I will provide a summary on my website / link to one
- discmath.ch contains a lot of good content
- We always cover the last week in a session (+ some of monday)

→ Everything you need to know for the exam is either in the script, or (linked) on my website, you don't need to take notes

My Website

Everything will be uploaded there
+ anything useful I have for you

Most content will be similar to
last year (will link to it)

Don't do the kahoots / exercises
before the session, we will work
on them in the session

Reading Questions

From last year, maybe they help
you

Plan for the sessions

First half: Kahoot + Theory recap

Break (~15mins, as needed)

Second half: exercises, tutorials
(solutions to exercises)

Learning by doing

I don't like to repeat stuff from the lecture.

Important that you ask questions.

Kahoots are designed to find misconceptions, don't worry if you get them wrong.

Later in semester = more theory
recaps (difficult content)

Feedback please (:

- **YOU** should benefit from this session, so make sure you do!
- NOTHING is fixed, tell me what you need, like, hate, ...
- Write me a mail, talk to me!

Okay, let's do Diskmath...

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Recap of Last Week

Abstraction

Important, but don't worry about it now.

You won't need to divide chocolate in the exam

Formulas and Statements

Statement

is true or false!

uses $\Rightarrow$, “and”, “or”, language

Formula

is like a function

uses $\rightarrow$, $\wedge$, $\vee$, A, B

evaluates to a truth value for an interpretation

When do I need to be exact?

When we do logic, you need to be exact.

Later on, something like “I like apples $\wedge$ I like trees” is fine

Now? **Be exact!**

Proving Equality of Formulas

Formula is small? → Compare truth tables for all cases

Longer Formula? → Use the rules from Lemma 2.1 to refactor it.

Be exact with the rules! Commutativity is also a rule, don't just assume it. Be precise for now, it's a good habit.

Exercise Sheets

You are allowed to use Lemmas from script only.

You cannot use anything we did not cover in the course.

Be precise, saying “this is true because it is” is not enough.

Kahoot

Theory

Exercises